

“Municipal Variations in Childhood Fitness and Excess Weight in Slovenia: A Panel Analysis”

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ABSTRACT

INTRODUCTION

Childhood physical fitness and excess weight are related but distinct dimensions of child health, and both may reflect the local environments in which children live, attend school, and participate in physical activity. National and regional averages can hide substantial variation across smaller geographic units. Slovenia offers an informative setting as the SLOfit surveillance system provides standardized, near-population measures of school-aged children, while municipality-level socioeconomic and demographic indicators are available annually.

METHODS

We construct a municipality-year panel from 2016 to 2025 that links the National Institute of Public Health (NIJZ) indicator of health-related physical fitness among children aged 6 to 15 with the municipality-level prevalence of excess weight and annual SiStat measures of population, average gross earnings, employment, municipal waste per capita, number of schools, number of enterprises, and net migration. The primary analysis estimates two-way fixed effects models at the municipality level and compares them with models aggregated to Slovenia's 12 statistical regions. A complementary descriptive analysis examines municipal correlates of childhood fitness.

RESULTS

At the region level, the association between childhood fitness and excess weight is negative but statistically insignificant ($\beta = -0.066$, $SE = 0.076$, $p = 0.406$). At the municipality level, the association is substantially larger and statistically significant both without controls ($\beta = -0.147$, $SE = 0.036$, $p < 0.01$) and after adjustment for observed municipality characteristics ($\beta = -0.127$, $SE = 0.036$, $p < 0.01$). The descriptive correlates analysis further indicates meaningful heterogeneity in the local characteristics associated with childhood fitness. Employment is positively associated with the fitness score, average gross earnings are negatively associated in the bivariate specification, and the remaining correlates are closer to zero. The accompanying analysis reports a joint F-test across the seven municipality characteristics of $F(7,209) = 8.94$ ($p < 0.001$).

DISCUSSION

The results show that municipality-level variation contains information that is obscured by broader regional aggregation. They further suggest that childhood fitness should be studied as a local population-health indicator in its own right rather than solely as a correlate of excess weight. As the analysis is observational and retroactive, the estimates are descriptive and should not be interpreted as individual level or causal effects. The findings motivate richer analyses of the socioeconomic, school, and built-environment conditions associated with local differences in childhood fitness.

1. Introduction

Childhood obesity remains major public health concerns as excess weight early in life is associated with adverse cardiometabolic risk, reduced physical functioning, poorer quality of life, and a greater likelihood of carrying excess weight into adulthood¹⁻⁴. At the same time, body mass index does not fully characterize the health and functional capacity of children. Physical fitness, particularly cardiorespiratory and muscular fitness, is increasingly recognized as an independent marker of current and future health. Higher physical fitness in childhood and adolescence is associated with more favorable cardiovascular risk profiles, and cardiorespiratory fitness has been described as an important clinical and population-health marker in youth³⁻⁴. This distinction is important for public health surveillance. Two communities with similar prevalence of excess childhood weight may nevertheless differ substantially in the functional capacity, activity opportunities, and health trajectories of their children.

Slovenia provides a distinctive study setting to study these patterns. The national SLOfit system has monitored the somatic and motor development of school-aged children for decades, generating broad and standardized population-level information on physical fitness and anthropometry⁵⁻⁹. Existing Slovenian studies show that the national story is not a simple one of uniform deterioration. Long-run analyses document improvements in several dimensions of physical fitness among more recent cohorts, particularly among girls, while also documenting increasing dispersion and polarization in fitness outcomes⁵⁻⁶. Similarly, the large rise in childhood overweight and obesity observed from the late 1980s through the 2000s was followed by a reversal or stabilization in many age-sex groups after 2010⁷. Nevertheless, these population averages do not imply that all children or all places experienced the same changes.

The possibility of geographic variation is especially relevant in a country with strong national education and physical activity institutions but substantial local variation in population size, economic conditions, school environments, transportation, and access to organized or informal physical activity. Slovenia historically has received high grades for school-based physical activity and has maintained extensive national physical fitness surveillance¹⁰⁻¹². However, the literature also emphasizes gaps in knowledge about community, and built-environment influences. Thus, a strong national institutional framework can coexist with local differences in the resources and environments that shape children's everyday opportunities to be active.

Recent evidence strengthens the case for looking below the national level. A 2026 study using WHO Childhood Obesity Surveillance Initiative data from five South-East European countries documented statistically significant subnational differences in childhood obesity within Slovenia and found that the Sub-national Human Development Index was inversely associated with childhood obesity across the pooled study regions¹³. In addition, related evidence from other countries also shows that physical fitness and physical activity can vary geographically with socioeconomic conditions, school infrastructure, and the built environment¹⁴⁻¹⁸. It suggests that broad regional averages may be too coarse when the relevant mechanisms operate through local characteristics such as schools, neighborhoods, recreation opportunities, employment conditions, migration.

This study therefore shifts attention from national and broad regional averages to the municipality. We collect annual data from 2016 to 2025 and ask two related questions. First, are within-area changes in municipality-level childhood fitness associated with changes in the prevalence of childhood excess weight, and does the estimated relationship differ when the same

data are aggregated to Slovenia's 12 statistical regions? Second, which observed municipality characteristics are descriptively associated with higher childhood fitness? The first question evaluates whether local fitness and excess weight move together over time and whether aggregation obscures the relationship. The latter treats childhood fitness as an outcome of interest and examines the local socioeconomic, demographic, and institutional correlates that may help explain municipal heterogeneity.

The paper makes three contributions. First, it provides a municipality-level perspective on childhood fitness and excess weight using a decade of annual Slovenian data. This is complementary to prior Slovenian studies that have focused primarily on national secular trends, individual-level differences, or broad subnational obesity patterns^{5-9,13}. In addition, the paper directly compares municipality- and region-level estimates, illustrating how geographic aggregation can change the signal observed in ecological public-health data. Lastly, it expands the empirical focus beyond the fitness and obesity relationship by examining the municipal correlates of childhood fitness. This extension responds directly to prior SLOfit research that identifies socioeconomic and environmental explanations for fitness inequality as an important area for future work⁵.

The main empirical strategy is two-way fixed effect (TWFE) and it is descriptive rather than causal. Municipality fixed effects remove time-invariant differences across places, and year fixed effects absorb shocks common to the country in a given year, but the remaining associations need not be causal. Changes in fitness, excess weight, employment, migration, and other municipality characteristics may be jointly determined by unobserved local changes. Moreover, the unit of observation is the municipality rather than the individual child, so the estimates should not be interpreted as individual-level behavioral effects. The goal is, instead, to

characterize where the strongest geographic associations appear and to identify patterns that can guide more detailed future work on local determinants of child health.

2. Background and Literature Review

2.1 Childhood Physical Fitness as a Population-Health Indicator

Physical activity and physical fitness are closely related but conceptually distinct. Physical activity describes behavior, a movement that increases energy expenditure, whereas physical fitness describes the capacity to perform physical tasks and reflects the combined influence of activity, growth and maturation, body composition, genetics, and other factors¹⁹. In youth, cardiorespiratory fitness and muscular fitness are particularly relevant as they are associated with cardiometabolic health and can track from childhood into later life^{3,4}. This distinction matters in surveillance research such that municipality-level fitness score is not simply a measure of how frequently children exercise. It is a functional outcome that reflects accumulated behavioral and biological processes.

Longitudinal evidence supports the importance of childhood fitness. A systematic review of longitudinal studies found that higher cardiorespiratory fitness in childhood and adolescence was associated with a more favorable future cardiovascular risk profile.⁴ The American Heart Association also emphasized cardiorespiratory fitness as an important marker of physical and mental health and academic achievement in youth.³ These studies motivate treating physical fitness not merely as an intermediate input into obesity prevention, but as a public health outcome with independent value.

Slovenia's SLOfit system provides a particularly strong empirical setting for population surveillance. A population-based study using 1,953,847 measurements from 26 Slovenian birth cohorts found positive secular trends in general physical fitness among recent cohorts,

particularly among girls, alongside increasing inequality and polarization in the fitness distribution.⁵ A subsequent population-based study of Slovenian children aged 7-15 documented heterogeneous long-term trends across components of physical fitness and between boys and girls.⁶ These studies demonstrate that average improvements can coexist with growing inequality, making local-level analysis especially relevant. If the children experiencing low fitness are geographically concentrated, national trends may conceal meaningful variation in local opportunity structures.

2.2 Childhood Excess Weight and Its Relationship with Fitness

Childhood excess weight is related to physical fitness through several pathways, but the relationship is not mechanical. Higher body mass can make weight-bearing fitness tests more difficult, while low physical activity can contribute simultaneously to lower fitness and positive energy balance. Diet, sleep, maturation, socioeconomic conditions, local environment can also influence both outcomes. Thus, physical fitness and excess weight should be regarded as correlated dimensions of health rather than interchangeable measures.

Slovenian evidence illustrates this point. A 30-year Slovenian surveillance study based on more than 6.7 million anthropometric observations documented a substantial rise in childhood overweight and obesity followed by reversals in many groups during the 2010s⁷. A population-wide cohort study found that children with different weight status experienced markedly different physical fitness trajectories during and after the COVID-19 pandemic⁸. A later 30-year surveillance study documented persistent and widening weight-based disparities in physical fitness⁹. Thus, the literature suggests a strong relationship between body weight and fitness while also showing that fitness contains additional information about functional health and resilience.

This study uses the NIJZ municipality indicator labeled “excessive nutrition” in the source documentation. Importantly, this measure is not an obesity-only outcome. It represents the share of children aged 6 - 15 whose BMI exceeds the age-specific threshold for overweight, and therefore includes both overweight and obesity. Thus, we refer to the empirical outcome as childhood excess weight or overweight and obesity prevalence. This terminology avoids overstating what the underlying indicator measures and aligns the manuscript with the official variable definition.²

2.3 Geographic and Socioeconomic Variation in Child Health

A growing literature emphasizes that childhood obesity and activity-related health outcomes are geographically varied. Place can matter through differences in household resources, school quality, access to sports and recreation facilities, transportation, neighborhood safety, green space, food environments, and social norms. These mechanisms may operate at spatial scales much smaller than a country or statistical region. An analysis that uses only national or region-level averages can therefore attenuate or average away local variation that is substantively important.

This concern is directly relevant to Slovenia. An analysis of 2019 COSI data from subnational regions in Croatia, Montenegro, North Macedonia, Serbia, and Slovenia found statistically significant regional heterogeneity in Slovenia and an inverse association between the Sub-national Human Development Index and childhood obesity across the pooled study regions.¹³ These findings demonstrate that even within relatively small countries, childhood obesity can vary meaningfully across place and socioeconomic context. The present analysis moves one geographic level further by examining municipalities and by exploiting annual variation over a ten-year period.

Recent fitness research also points toward meaningful regional differences. A 2026 study of 42,741 Slovak schoolchildren found that regional variation in physical fitness was associated with BMI, life expectancy, socioeconomic characteristics, and school infrastructure, with substantial differences by sex and age group.¹⁸ Although institutional settings differ across countries, the study reinforces a general conclusion that fitness is not only an individual trait but can also reflect the broader systems and environments where children develop.

2.4 Local Economic, School, and Built-Environment Correlates

The determinants of childhood physical activity and fitness are multidimensional. Household and area-level socioeconomic resources may affect access to organized sports, equipment, transportation, and safe spaces for activity. At the same time, the relationship between socioeconomic position and youth activity is not uniform across settings or measures. U.S. national survey data show socioeconomic gradients across several youth fitness measures, while broader reviews indicate that the evidence linking socioeconomic position to young people's physical activity is mixed and sensitive to the socioeconomic indicator, activity domain, and age group examined.¹⁵⁻¹⁷ This mixed evidence cautions against interpreting a single municipal economic variable as a comprehensive measure of health opportunity.

The built environment is another plausible source of local variation. Reviews of high-income countries suggest that walkability, parks and recreation facilities, transportation infrastructure, and related neighborhood features are associated with children's physical activity, while also emphasizing heterogeneity in study quality and findings.¹⁴ Slovenia's own physical activity report cards consistently highlight the central role of schools, organized sport, government policy, and local infrastructure¹⁰⁻¹². For example, the 2016 report gave a high grade to schools but explicitly called for more evidence on family and built-environment influences.¹⁰

These mechanisms motivate the municipality characteristics examined in this study. Population captures demographic scale such that average gross earnings and employment capture distinct dimensions of the local economic environment, the number of schools proxies educational infrastructure, the number of enterprises captures local economic activity, net migration reflects demographic dynamism and possible sorting across places, and municipal waste per capita is included as an available municipality-level measure related to local consumption and service conditions. None of these variables should be treated as a complete measure of the local environment. Rather, together they provide an initial empirical description of the types of municipalities associated with higher childhood fitness and allow the main fitness and excessweight association to be evaluated conditional on observed local characteristics.

2.5 Study Expectations and Empirical Framing

The literature does not imply a single prediction for every municipality characteristic. We expect childhood fitness and excess weight to be inversely associated on average, but the magnitude of this association may depend on geographic scale. If important variation occurs within regions, aggregation should attenuate the relationship and reduce statistical power. For the correlates analysis, stronger local employment conditions and greater school or economic capacity could plausibly be associated with higher fitness, but higher earnings may also correlate with urbanization, time constraints, sedentary lifestyles, or other unmeasured characteristics. Migration and population size may similarly reflect multiple competing mechanisms. The correlates analysis is therefore presented as exploratory and descriptive rather than as a test of a tightly parameterized causal model.

3. Methods

3.1 Data Sources and Study Sample

The analysis combines municipality-level data from the National Institute of Public Health (NIJZ), the SLOfit system, and the Statistical Office of the Republic of Slovenia (SiStat). NIJZ's "Health in the Municipality" indicator system provides annual municipality-level measures of health-related childhood physical fitness and childhood excess weight. The underlying childhood measures are derived from SLOfit school surveillance. SiStat provides annual demographic, labor market, economic, and municipal characteristics used as covariates and as predictors in the descriptive fitness analysis. The study period is encompassing 2016 through 2025, corresponding to the annual availability of the NIJZ municipality indicators used here^{2,21,22}.

The main unit of observation is the municipality-year. The merged panel contains 211 municipalities. The reported municipality regressions contain 2,012 observations without controls and 1,797 observations with the full control set, with 210 municipality clusters in both specifications. For comparison, municipality-level fitness and excess-weight measures are aggregated to the 12 Slovenian statistical regions by year, producing 120 region-year observations.

The use of repeated annual municipality data allows the primary specification to focus on within-place changes over time rather than cross-sectional differences alone. This is useful as municipalities differ persistently in geography, culture, infrastructure, and population composition. Municipality fixed effects absorb these time-invariant differences. Nevertheless, the analysis remains aggregate as the unit is an area-year average, and children are not linked individually across outcomes or municipality characteristics.

3.2 Childhood Physical Fitness Measure

The main explanatory variable in the fitness and excess-weight association analysis is NIJZ indicator 2.1, health-related physical fitness of primary-school children and adolescents aged 6 - 15. The indicator describes the municipality-level motor efficiency relevant to health. It is calculated from four physical-fitness components including muscular endurance, aerobic endurance, muscular strength, and flexibility. The underlying SLOfit tasks used by NIJZ are sit-ups, the 600-meter run, sit-and-reach, and bent-arm hang.²

The component measures are standardized by age and sex. NIJZ then summarizes performance such that the Slovenian average corresponds to the 50th percentile. Municipality averages above 50 indicate above-average motor efficiency relative to the national reference, while values below 50 indicate below-average fitness. Accordingly, the variable should be interpreted as a municipality-level fitness score or index, not as the prevalence of children who are “fit”.

Coverage of the underlying SLOfit measurements is a major strength. According to the NIJZ indicator documentation, measurements used to construct the municipality index cover approximately 96% of primary-school children aged 6 - 15. Measurements are conducted for children in municipalities with primary schools. For municipalities without a primary-school seat, the index can be assigned based on the municipality containing the school serving that school catchment area. Daily school absence means the data are not literally complete, but coverage is unusually high for population fitness surveillance.²

3.3 Childhood Excess-Weight Measure

The outcome in the main regression is NIJZ indicator 2.2, the share of primary-school children and adolescents aged 6 - 15 whose body mass index exceeds the age-specific threshold for overweight. The indicator therefore includes all children classified as overweight or obese

rather than obesity alone. BMI is calculated from measured height and weight collected through SLOfit. As BMI changes systematically with age during childhood, the NIJZ documentation relies on age-specific cut points aligned with the International Obesity Task Force approach^{2,20}.

As with the fitness indicator, the underlying measurements cover approximately 96% of primary-school children. The result is a municipality-level percentage. A one-unit change in the dependent variable corresponds to a one-percentage-point change in the share of measured children above the age-specific excess-weight threshold. Throughout the empirical sections, we use *excess weight* or *overweight/obesity* prevalence to reflect the content of this indicator precisely.

3.4 Municipality-Level Covariates

The adjusted municipality specification includes seven time-varying characteristics from SiStat including total population, average gross earnings in euros, employment rate, municipal waste per capita, number of schools, number of enterprises, and net migration per 1,000 residents. These variables were selected as they provide annual coverage and represent several dimensions of the local environment that may co-vary with childhood health. Table 1 summarizes each variable, its empirical role, and the conceptual domain it captures.

Population is included to account for changes in municipal scale and density-related pressures. Average gross earnings measure local labor market compensation and are treated as an economic resource indicator. Employment rate captures labor market condition and may reflect both household economic security and broader local economic conditions. Municipal waste per capita is an available proxy for local consumption and municipal service conditions. The number of schools captures educational infrastructure but it does not measure school quality, physical education intensity, or sports facilities directly. The number of enterprises captures local

economic activity. Net migration per 1,000 residents measures annual population inflows relative to outflows and may reflect local opportunity, residential sorting, and demographic change.

3.5 Statistical Analysis

We begin with a region-level two-way fixed-effects (TWFE) model. Let r index statistical region and t index year. We estimate following equation:

$$Excessweight_{rt} = \beta Fitness_{rt} + \mu_r + \lambda_t + \epsilon_{rt}$$

Region fixed effect (μ_r) absorb time-invariant differences across the 12 regions, and year fixed effect (λ_t) absorb national shocks common to all regions in a year. Standard errors are clustered at the region level. As there are only 12 regional clusters, inference is imprecise and the region-level model is treated as a coarse comparison rather than the main specification.

The primary municipality-level model is:

$$Excessweight_{mt} = \beta Fitness_{mt} + \mu_m + \lambda_t + \epsilon_{mt}$$

where m indexes municipality and t indexes year. Municipality fixed effect (μ_m) absorb time-invariant local characteristics and year fixed effect (λ_t) absorb common national changes. Standard errors are clustered by municipality. The identifying variation in this specification comes from changes in the municipality fitness score over time relative to changes in excess weight in the same municipality, net of national year effects.

We then add the seven observed municipality characteristics in the following equation,

$$Excessweight_{mt} = \beta Fitness_{mt} + X\Gamma + \mu_m + \lambda_t + \epsilon_{mt}$$

where X contains population, average gross earnings, employment rate, municipal waste per capita, number of schools, number of enterprises, and net migration per 1,000 residents. Comparing β across the unadjusted and adjusted models indicates how sensitive the fitness and excess-weight association is to the observed time-varying local characteristics.

Finally, we conduct a complementary descriptive analysis of municipality characteristics associated with childhood fitness. For each characteristic Z_{mt} , the coefficient plot (Figure3) reports a bivariate linear OLS regression of the following form,

$$Fitness_{mt} = \delta Z_{mt} + u_{mt}$$

Variables in the coefficient plot are rescaled to make magnitudes interpretable. Average gross earnings are expressed per €1,000, population per 10,000 residents, employment rate per 10 percentage points, municipal waste per 100 kg per capita, number of schools per 10 schools, number of enterprises per 10,000 enterprises, and net migration per 10 migrants per 1,000 residents. These coefficients are descriptive correlations, not causal effects. The accompanying analysis reports that the seven municipality characteristics are jointly informative for childhood fitness, with $F(7,209) = 8.94$ ($p < 0.001$).

4. Results

4.1 Descriptive Geographic Variation

Figures 1 and 2 show the 2025 municipality distributions of the childhood fitness score and childhood excess weight prevalence. The maps show substantial geographic heterogeneity in both outcomes. Municipalities with relatively high fitness scores are not concentrated uniformly in a single part of the country, and excess weight prevalence varies substantially across local areas. The spatial patterns provide a visual motivation for using municipalities rather than relying solely on national or regional averages.

The maps should be interpreted descriptively rather than as evidence of a causal spatial mechanism. They summarize one year and do not account for population size, timing variation, or correlated local characteristics. The primary goal is to show that meaningful heterogeneity

exists at a geographic scale below the 12 statistical regions and to motivate the panel regressions that follow.

4.2 Region-Level Association Between Fitness and Excess Weight

Table 2 reports the region-level two-way fixed-effects (TWFE) model. The estimated coefficient on childhood fitness is -0.066 (SE = 0.076, $p = 0.406$). The point estimate has the expected negative sign AS higher regional fitness is associated with lower regional excess-weight prevalence, but it is not statistically significant. The model includes region and year fixed effects, contains 120 region-year observations, and clusters standard errors across 12 regions.

The lack of precision is consistent with two limitations of regional aggregation. First, averaging municipality values within regions reduces the amount of local variation available for estimation. If municipal changes in fitness and excess weight are heterogeneous within a region, the region average can cancel or dilute that variation. Further, inference relies on only 12 regional clusters so the resulting estimate should therefore be interpreted as a coarse benchmark rather than evidence that the fitness and excess-weight relationship is absent.

4.3 Municipality-Level Association Between Fitness and Excess Weight

Table 3 presents the primary municipality-level estimates. In the baseline municipality fixed-effects model without additional controls, the coefficient on the childhood fitness score is -0.147 (SE = 0.036, $p < 0.01$). Thus, within municipalities, years with a one point higher fitness score are associated with approximately a 0.147 percentage point lower prevalence of excess weight, relative to common year changes. The estimate is substantially larger in magnitude and more precise than the region-level coefficient.

Column (2) in Table3 adds population, average gross earnings, employment rate, municipal waste per capita, number of schools, number of enterprises, and net migration per

1,000 residents. The childhood fitness coefficient remains negative and statistically significant at -0.127 (SE = 0.036, $p < 0.01$). The modest attenuation from -0.147 to -0.127 suggests that the observed municipality characteristics explain part, but not most, of the within-municipality co-movement between fitness and excess weight.

The comparison across geographic scales is one of the central empirical results. The direction of the association is negative at both levels, but the municipality estimate is more than twice as large in absolute magnitude and is estimated with substantially greater precision. This pattern is consistent with the idea that broader aggregation obscures meaningful local heterogeneity. It also reinforces the value of municipality-level surveillance for detecting relationships that may not be visible in region-level averages.

4.4 Municipality Characteristics Associated with Childhood Fitness

Figure 3 turns the empirical question around and treats childhood fitness as the outcome. The figure reports descriptive bivariate associations between the municipality fitness score and each of the seven local characteristics, with variables rescaled for readability. The pattern shows that local conditions are not uniformly associated with fitness in the same direction.

The largest positive association in the coefficient plot is for the employment rate. A 10-percentage-point increase in employment is associated with a noticeably higher municipality fitness score. Average gross earnings, by contrast, show the largest negative bivariate association. This contrast is substantively useful as it cautions against treating municipal socioeconomic status as a one-dimensional construct. Employment may proxy household stability, labor-market attachment, or broader community functioning, whereas earnings may covary with urbanization, occupational structure, housing costs, parental time constraints, or other characteristics not observed in the current data.

Population is estimated close to zero. Municipal waste, the number of schools, the number of enterprises, and net migration show smaller positive associations, with confidence intervals that are visibly wider relative to the central estimate for several variables. The accompanying joint test indicates that the set of seven municipal characteristics is collectively associated with childhood fitness ($F(7,209) = 8.94$, $p < 0.001$), even though individual coefficients differ substantially in magnitude and precision.

These results should be read as descriptive correlates rather than “predictors” in a causal sense. In particular, the bivariate coefficients can reflect omitted differences across municipalities, and the observed sign can change when variables are entered jointly or when fixed effects are added. Nevertheless, the figure serves an important role as it demonstrates that childhood fitness itself contains meaningful local socioeconomic variation and motivates expanding the municipality data beyond the seven currently available annual characteristics.

4.5 Multivariable Municipal Correlates of Excess Weight

The controlled specification in Table 3 also provides information on the measured municipal correlates of excess weight conditional on childhood fitness and fixed effects. The coefficients on population (-0.0001), average gross earnings (0.001), employment rate (-0.058), municipal waste per capita (-0.0003), number of schools (-0.499), number of enterprises (0.0003), and net migration (-0.004) are small relative to their standard errors in the reported model. None is statistically significant.

This lack of significance does not imply that local socioeconomic or institutional conditions are unimportant. The fixed-effects design uses only within-municipality changes, and many structural characteristics of municipalities change slowly. Measurement error in broad proxies such as number of schools or enterprises can further attenuate coefficients. The result is

better interpreted as evidence that the childhood fitness score is the most stable measured correlate of excess weight in the current within-municipality specification, while richer measures of the local environment are needed to characterize mechanisms.

5. Discussion

This study uses a decade of municipality-level Slovenian data to examine the local relationship between childhood physical fitness and excess weight and to characterize municipal correlates of childhood fitness. Three findings stand out. First, the fitness and excess-weight association is negative at both region and municipality levels but is statistically significant only at the municipality level. Second, the municipality estimate remains sizable after adjustment for observed socioeconomic, demographic, and municipal characteristics. Third, childhood fitness itself is systematically related to local conditions, with particularly strong descriptive associations for employment and average gross earnings in opposite directions.

The contrast between region- and municipality-level estimates is crucial. Regional averages are useful for broad surveillance and policy comparison, but they can conceal variation among local areas that differ in schools, population composition, economic activity, transport, recreation opportunities, and social organization. The 2026 COSI evidence of subnational obesity heterogeneity within Slovenia shows that geographic inequality exists even at the regional scale¹³. The results emphasize that substantial additional variation operates within those regions. This is consistent with the broader literature showing that local environments can matter for physical activity, fitness, and obesity^{14,18}.

The findings also support a broader interpretation of physical fitness as a population-health indicator. Much of the obesity literature treats physical activity or fitness primarily as a mechanism for weight control. Yet physical fitness is associated with health independently of

weight status and captures functional capacity that BMI cannot^{3,4}. The Slovenian data are particularly well suited for this perspective as the fitness indicator is based on standardized school measurements with very high coverage. Monitoring municipality-level fitness can therefore complement weight surveillance and may help identify areas where children have limited functional capacity even when excess-weight prevalence alone does not appear unusually high.

The negative municipality-level association between fitness and excess weight is consistent with prior Slovenian research showing substantial weight-based disparities in physical fitness^{8,9}. However, the aggregate coefficient should not be interpreted as the effect of increasing an individual child's fitness. Municipalities experiencing improvements in fitness may simultaneously experience changes in school programs, sports participation, family behavior, food environments, population composition, or other determinants that also affect excess weight. The coefficient summarizes area level co-movement rather than an individual treatment effect.

The correlates analysis adds a second substantive contribution. Prior SLOfit work documents growing polarization in physical fitness and explicitly calls for research on socioeconomic covariates and the unequal opportunity structures underlying fitness development⁵. Our municipal analysis is an initial response to that gap. The employment coefficient suggests that local labor-market conditions may be related to children's physical fitness, but the negative bivariate coefficient on average gross earnings makes clear that simple "richer is healthier" narratives are insufficient. Area-level earnings and employment can capture different local processes, and both may be correlated with urbanicity, occupational structure, commuting, family time allocation, sports access, and residential sorting.

The school variable likewise requires cautious interpretation. Slovenia has a strong national physical education system and has historically received high marks for the school setting¹⁰⁻¹². Thus, counting schools within a municipality is a coarse proxy for child specific access or school quality. It does not measure physical education hours, specialist staffing, gymnasiums, outdoor facilities, travel distance, and adherence or dropouts. A weak coefficient on number of schools should not be interpreted as evidence that the school environment is irrelevant. Future work would benefit from linking the municipality panel to measures of school sports infrastructure, PE intensity, organized sport participation.

The same interpretation applies to the built environment. The current annual panel does not include direct measures of playground access, recreation facilities, green space, safe walking and cycling routes, sports-facility density, or neighborhood cohesion. NIJZ's broader municipal indicator system contains several conceptually relevant measures including access to recreation surfaces, access to safe walking and cycling paths, and municipality development but some are survey-based and available only periodically rather than annually. These measures are promising natural extensions for cross-sectional analyses, but they should not be put into the annual fixed-effects model without careful attention to temporal comparability.

There are several limitations in this study. First, the study is aggregate level analysis. Area-level associations cannot be translated into individual level causal effects. In addition, reverse causality and simultaneity are plausible as fitness and excess weight may evolve jointly in response to the same local changes. Third, the available annual covariates are broad proxies and do not capture many mechanisms emphasized in the literature.. Lastly, the four-component NIJZ fitness indicator is designed as a health related local indicator and differs from broader

eight-test composite fitness measures used in some SLOfit research. Thus, comparisons across studies should recognize differences in construction.

Despite these limitations, the study has notable strengths. It uses standardized measures with near-population coverage, a ten-year panel, a fine geographic unit, and a design that explicitly distinguishes local from broader regional variation. The municipality-year structure also creates a platform that can be enriched as additional socioeconomic, school, environmental, and policy variables become available. Thus, results are useful not only as a description of the fitness and excess weight relationship but also as a foundation for a broader research agenda on local determinants of childhood functional health.

From a policy perspective, the key implication is not that municipalities with a particular observed characteristic should immediately be targeted based on a single coefficient. Rather, the findings support the value of local surveillance and local diagnosis. If municipality-level fitness contains information hidden by regional averages, then national prevention strategies may benefit from identifying local areas with persistently low fitness, high excess weight, or worsening trajectories and then investigating the specific school, infrastructure, socioeconomic, and behavioral mechanisms present in those places. This approach is consistent with Slovenia's strong tradition of population fitness monitoring while extending it toward place-based public-health planning.

6. Conclusion

Childhood physical fitness and excess weight vary substantially across Slovenian municipalities. In region-level panel models, the association between fitness and excess weight is negative but statistically imprecise. At the municipality level, the relationship is considerably stronger. Higher childhood fitness scores are associated with lower excess-weight prevalence,

and the association remains statistically significant after adjustment for observed municipality characteristics. The difference across geographic scales indicates that local variation can be obscured when municipalities are aggregated into broader regions.

The complementary correlates analysis further shows that childhood fitness is itself geographically and socioeconomically patterned. Employment is positively associated with the fitness score in the descriptive bivariate analysis, average gross earnings are negatively associated, and the remaining municipal characteristics exhibit smaller relationships. These patterns underscore the multidimensional nature of local socioeconomic context and argue against interpreting childhood fitness solely through the lens of body weight.

The main contribution of the study is therefore twofold. It documents a robust municipality-level relationship between childhood fitness and excess weight, and it reframes childhood fitness as a local population-health outcome whose determinants merit direct study. Future work should expand the municipality panel with more precise measures of school resources, sports and recreation infrastructure, transportation, green space, social conditions, and child specific composition, while retaining the high coverage longitudinal surveillance that makes Slovenia a valuable setting for studying the geography of childhood health and fitness.

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Tables and Figures

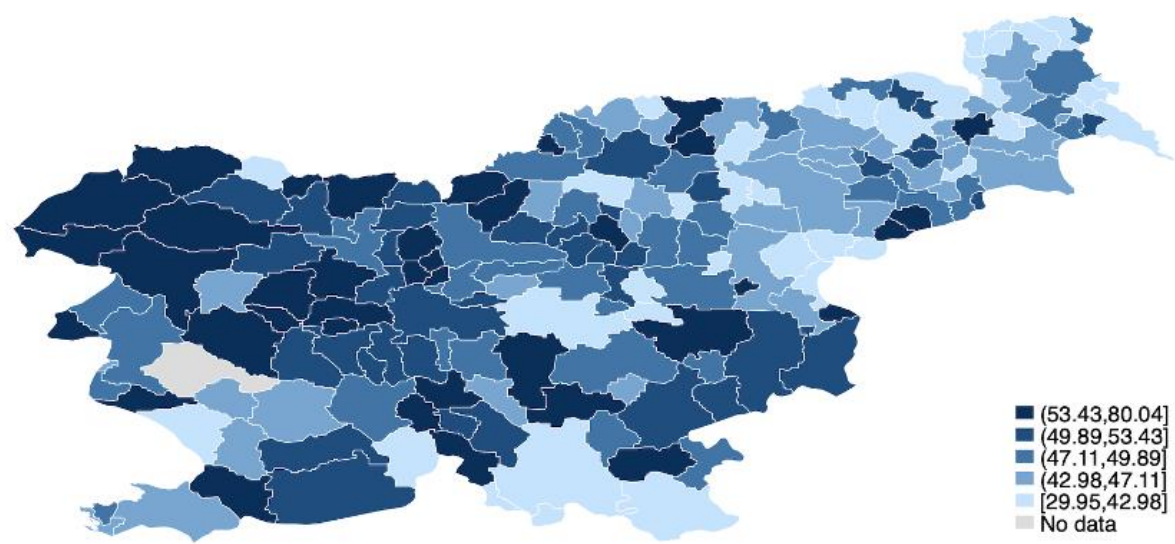


Figure 1 : Childhood Fitness Score by Municipality, 2025

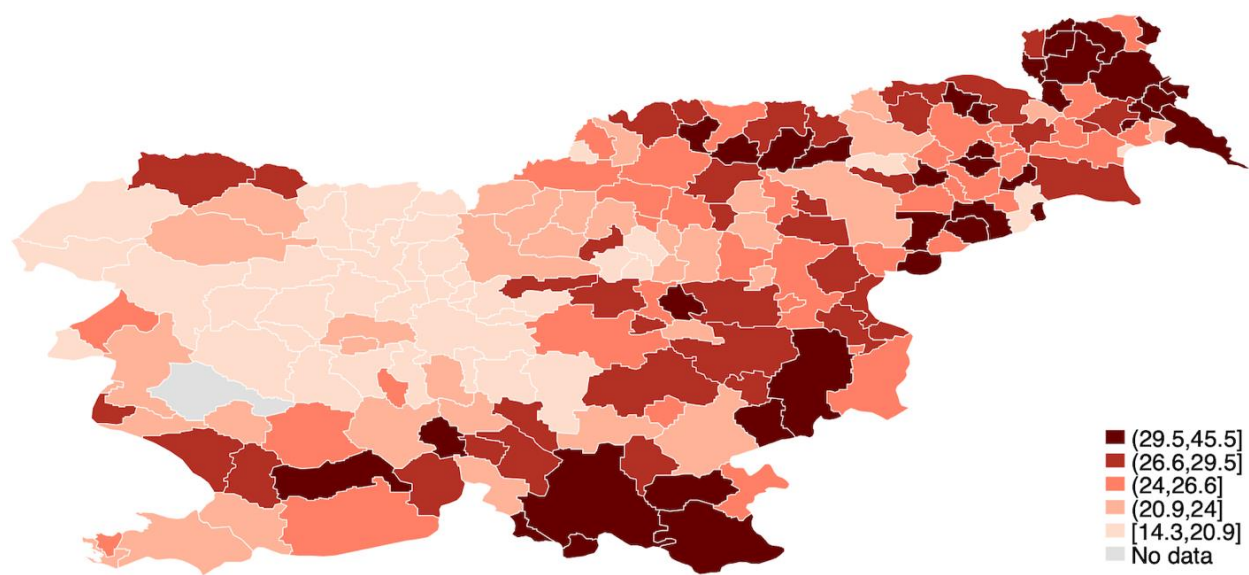


Figure 2 : Childhood Excess-Weight Prevalence by Municipality, 2025

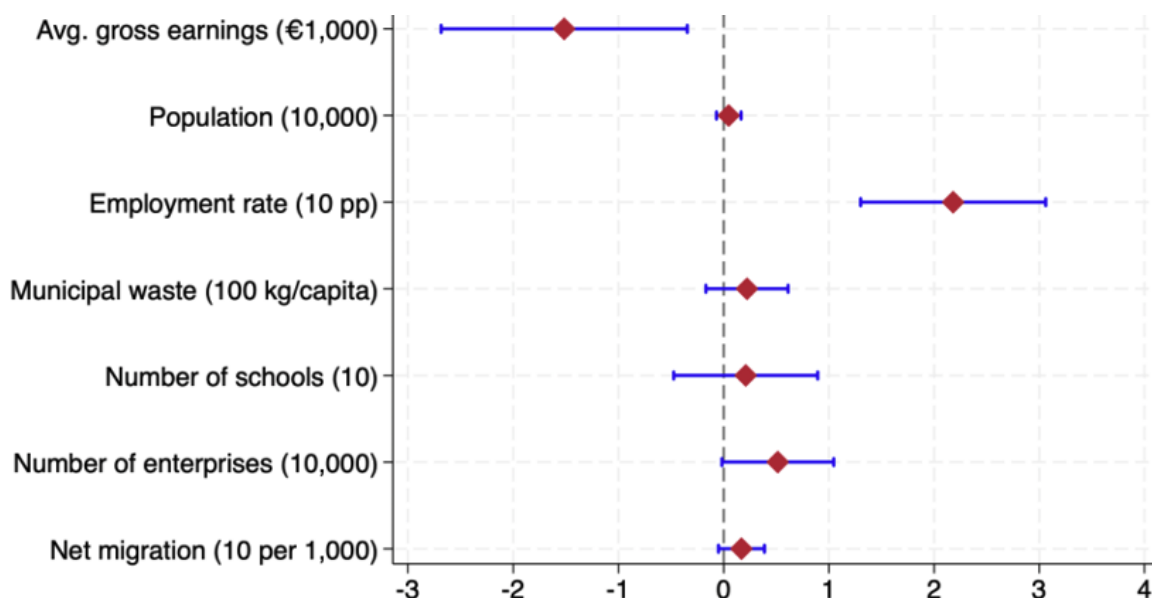


Figure 3 : Descriptive Correlates of Childhood Fitness

Note for Figure 3: Points report descriptive bivariate associations between the childhood fitness score and each municipality characteristic for 2016–2025. Variables are rescaled as shown on the y-axis. Error bars show the reported confidence intervals. The accompanying analysis reports a joint F-test across the seven characteristics of $F(7,209) = 8.94$, $p < 0.001$. The estimates are descriptive and should not be interpreted causally.

Table 1. Municipality-Level Variables, Definitions

Variable	Definition	Unit	Source
Childhood fitness score	Health-related motor-fitness index for children ages 6–15 based on muscular endurance, aerobic endurance, muscular strength, and flexibility; national average = 50.	Index / percentile-like score	NIJZ / SLOfit
Childhood excess weight	Share of measured children ages 6–15 with BMI above the age-specific overweight threshold; includes overweight and obesity.	Percent	NIJZ / SLOfit
Population	Total municipality population as reported by SiStat.	Residents	SiStat
Average gross earnings	Average gross earnings in euros as reported by SiStat.	EUR	SiStat
Employment rate	Municipality employment rate.	Percent	SiStat
Municipal waste per capita	Municipal waste generated per resident.	kg per capita	SiStat
Number of schools	Count of schools in the municipality.	Count	SiStat
Number of enterprises	Count of enterprises in the municipality.	Count	SiStat
Net migration	Net migration relative to municipality population.	Per 1,000 residents	SiStat

Notes: The study period is 2016–2025. The NIJZ childhood fitness indicator is a score, not a prevalence measure. The excess-weight indicator includes children above the age-specific overweight threshold and therefore combines overweight and obesity. In Figure 3, correlates are rescaled for presentation.

Table 2. Summary Statistics Across Slovenia's 12 Statistical Regions, 2016–2025									
Statistical region	Childhood outcomes		Socioeconomic characteristics			Municipal characteristics			
	Fitness	Obesity	Population	Avg. gross earnings (€)	Employment rate (%)	Waste (kg/capita)	Schools	Enterprises	Net migration (per 1,000)
Gorenjska	53.39	21.55	207,442	1,850.04	69.41	332.68	78.10	19,091.80	4.23
Goriška	51.32	22.33	98,539	1,775.99	67.44	403.74	55.40	9,351.60	1.97
Jugovzhodna Slovenija	48.76	28.23	145,321	1,743.44	66.93	271.52	63.90	10,081.70	4.27
Koroška	47.77	29.19	70,647	1,789.16	61.37	271.57	37.20	5,152.60	-0.85
Obalno-kraška	48.57	25.56	113,359	1,764.89	65.92	390.41	40.00	12,941.50	9.65
Osrednjeslovenska	51.00	22.76	553,084	1,816.09	69.50	330.17	163.00	64,429.50	7.54
Podravska	47.36	30.10	321,028	1,692.65	62.17	281.39	112.00	25,048.90	5.03
Pomurska	42.70	33.48	114,475	1,669.23	56.15	285.51	53.60	7,372.00	2.12
Posavska	48.78	28.95	75,776	1,763.03	66.61	311.13	31.00	5,431.80	1.59
Primorsko-notranjska	49.20	27.13	53,110	1,685.85	70.49	319.43	27.00	4,244.10	2.40
Savinjska	49.14	27.31	258,038	1,697.41	67.70	314.77	114.30	20,882.60	5.46
Zasavska	45.12	28.78	57,137	1,782.82	64.83	325.94	28.00	3,675.10	2.28

Table 2. Relationship Between Childhood Fitness and Excess Weight: Region-Level Fixed-Effects Estimate

	Excess weight
Fitness	-0.066 (0.076)
Region fixed effects	Yes
Year fixed effects	Yes
Observations	120
Clusters	12
Adjusted R-squared	0.957

Notes: The dependent variable is childhood excess-weight prevalence among children ages 6–15. The main explanatory variable is the childhood fitness score. The model uses 12 Slovenian statistical regions observed from 2016 to 2025 and includes region and year fixed effects. Standard errors clustered at the region level are in parentheses. *** $p < 0.01$.

Table 3. Relationship Between Childhood Fitness and Excess Weight by Municipality

	(1) Excess weight	(2) Excess weight
Fitness	-0.147*** (0.036)	-0.127*** (0.036)
Population		-0.0001 (0.0003)
Average gross earnings (EUR)		0.001 (0.002)
Employment rate		-0.058 (0.097)
Municipal waste per capita		-0.0003 (0.001)
Number of schools		-0.499 (0.458)
Number of enterprises		0.0003 (0.0004)
Net migration per 1,000		-0.004 (0.004)
Municipality fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Municipality-level controls	No	Yes
Observations	2,012	1,797
Clusters	210	210
R-squared	0.800	0.809
Adjusted R-squared	0.776	0.782
Within R-squared	0.035	0.028

Notes: The dependent variable is childhood excess-weight prevalence. Fitness is the municipality-level childhood fitness score. Both specifications include municipality and year fixed effects. Column (2) adds population, average gross earnings, employment rate, municipal waste per capita, number of schools, number of enterprises, and net migration per 1,000 residents. Standard errors clustered at the municipality level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.